Event-Driven Agentic SOC (ED-ASOC): Supervisor-Based LLM Framework for Dynamic Incident Response and SOAR Orchestration

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Abstract—As cyber threats grow more sophisticated, Security Operations Centers (SOC) face mounting pressure from the gap between alert volumes and the speed of traditional incident response. Analysts struggle with cognitive overload caused by high false-positive rates, leading to alert fatigue and missed critical threats. Current SOAR platforms, despite automating routine tasks, rely on polling-based data retrieval and static playbooks that cannot adapt to polymorphic or zero-day attacks. This paper presents the Event-Driven Agentic SOC (ED-ASOC) framework, built around a Supervisor Agent that enables the shift from scripted automation to autonomous orchestration. By replacing periodic polling with Webhook-based real-time ingestion, the framework cuts detection-to-analysis latency from minutes to sub-second levels. The Supervisor Agent enriches raw telemetry with organizational context through Retrieval-Augmented Generation (RAG), producing adaptive responses that move beyond rigid rules. The agent can execute remediation actions, including IP blocking, endpoint isolation, and user suspension, via defined API schemas, closing the gap between detection and containment. A tiered Human-in-the-Loop (HITL) mechanism ensures that high-impact decisions require analyst approval. Experimental evaluation using FortiSIEM and FortiSOAR in a controlled testbed demonstrates a 73% reduction in Mean Time to Response (MTTR) and 89% decrease in detection latency compared to polling-based baselines, while reducing Tier-1 analyst workload by approximately 45%.

Keywords—Agentic AI, Security Operations Center, SOAR, Large Language Models, Incident Response, Retrieval-Augmented Generation, Digital Forensics, Human-in-the-Loop

I. Introduction

Cybersecurity operations have become a critical requirement as adversaries increasingly use artificial intelligence and automation to compress attack lifecycles. Security Operations Centers (SOC) now face severe structural constraints. The rapid growth in telemetry data, combined with complex multi-stage attack patterns, traps analysts in continuous triage cycles. Alert fatigue (where notification volume desensitizes analysts) often causes delayed responses or missed legitimate threats [1, 2]. Traditional SOAR solutions attempt to reduce this burden but depend on rule-based correlation and static playbooks. These deterministic systems use rigid conditional logic, making them ineffective against polymorphic malware, zero-day exploits, or living-off-the-land techniques that bypass known signatures.

A core limitation of current SOC architecture lies in data integration. Most SIEM integrations use periodic polling (e.g., every 60 seconds). For modern ransomware or advanced persistent threats (APTs), where lateral movement can occur within minutes of initial compromise, this polling delay creates a critical vulnerability window. Additionally, current automated systems lack reasoning capabilities; they execute scripts but cannot infer intent or connect disparate events without explicit programming.

Despite widespread AI adoption in security, most implementations remain passive. They assist with query generation or log summarization but cannot interact with the environment. This dependency on human intervention for every action creates bottlenecks that negate AI's speed advantages. A truly autonomous system must detect and act, using integrated tools to contain threats before they spread. This capability, termed Agentic AI, represents the next step in defensive cyber operations.

This paper proposes the Event-Driven Agentic SOC (ED-ASOC) architecture, designed to enable the transition from automated security (static scripts) to autonomous security (dynamic, goal-oriented orchestration). The core component is the Supervisor Agent, an autonomous decision engine powered by Large Language Models (LLM). Unlike passive analysis tools that only summarize data, this agent has actionable intelligence, enabling direct interaction with infrastructure components (firewalls, EDR agents, and identity providers) to neutralize threats in real time [3, 4].

Through Retrieval-Augmented Generation (RAG), the system captures successful incident response patterns beyond individual analyst expertise, storing organizational knowledge in a retrievable vector store. This ensures every

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response draws from the collective history of past incidents and verified security policies.

The primary contributions of this study are:

1) Event-Driven Architecture: A Webhook-based ingestion layer that eliminates polling latency, achieving sub-second threat detection [5].
2) Contextual Intelligence: A RAG-driven decision framework enabling the Supervisor Agent to generate dynamic response plans adapted to specific threat contexts [6, 7].
3) Actionable Autonomy: A structured Tool Calling mechanism that empowers AI to autonomously execute remediation commands on critical infrastructure [8].
4) Explainable Auditability: A Human-in-the-Loop (HITL) protocol that categorizes actions by risk, requiring human authorization for high-impact decisions [9–11].

II. Related Work

A. Traditional SOC Automation and SOAR Platforms

Security Orchestration, Automation, and Response (SOAR) platforms emerged to address the operational challenges of modern SOCs. Bridges et al. [8] conducted extensive testing of commercial SOAR tools, finding that while these platforms effectively automate repetitive tasks, they struggle with novel threat patterns due to their reliance on predefined playbooks. Islam et al. [12] proposed SmartValidator for automatic threat classification, demonstrating improvements over manual triage but noting limitations in handling context-dependent decisions.

B. Large Language Models in Cybersecurity

Recent advances in LLMs have opened new possibilities for security applications. Aghaei et al. [13] introduced SecureBERT, a domain-specific language model trained on cybersecurity corpora. Ji et al. [14] presented SEvenLLM, benchmarking LLM capabilities in cyber threat intelligence. Habibzadeh and Feyzi [2] provided a comprehensive survey of LLM applications in SOCs, identifying alert triage and incident summarization as primary use cases. However, these implementations primarily use LLMs for text processing rather than autonomous action.

C. Agentic AI and Autonomous Systems

The concept of Agentic AI, referring to systems capable of independent reasoning and action, has gained attention in recent literature. Wu et al. [3] proposed frameworks for enhancing LLM reasoning through agentic tools. Derouiche et al. [4] analyzed architectural patterns and design challenges for agentic systems. Lazer et al. [15] surveyed agentic AI applications in cybersecurity, identifying autonomous incident response as a promising but underexplored area. These works establish theoretical foundations but lack practical implementations integrating agentic capabilities with production SOAR platforms.

D. Research Gap

While prior work has explored LLMs for security text analysis and theory about agentic architectures, no existing framework combines: (1) event-driven real-time ingestion, (2) RAG-enhanced contextual reasoning, (3) autonomous tool execution via SOAR APIs, and (4) tiered HITL governance, all of which are integrated into a deployable architecture with empirical validation. ED-ASOC addresses this gap.

III. System Architecture

The ED-ASOC framework comprises five interconnected components orchestrated through an event-driven message bus, as illustrated in Fig. 1. The architecture follows a layered design pattern separating concerns across ingestion, reasoning, knowledge management, execution, and governance layers. This modular approach enables independent scaling of components based on operational load while maintaining loose coupling through well-defined interfaces.

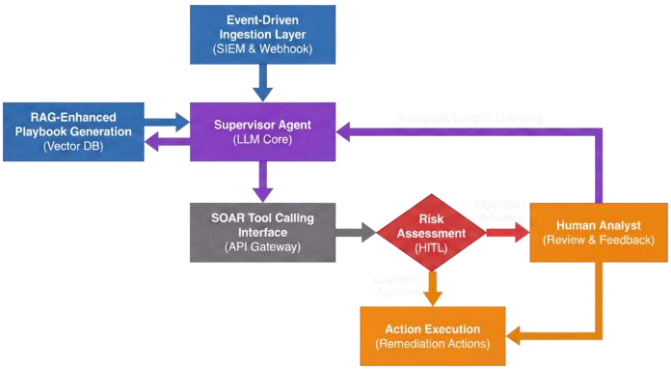


Fig. 1. ED-ASOC System Architecture showing event-driven ingestion, Supervisor Agent with LLM core, RAG-enhanced playbook generation, SOAR tool calling interface, risk assessment with HITL mechanism, and action execution flow.

A. Event-Driven Ingestion Layer

Conventional SOCs employ periodic polling strategies (e.g., every 30-60 seconds) to retrieve alerts. This polling interval introduces both resource overhead and detection latency. The proposed Webhook layer uses a push-based model [5]. Regardless of the SIEM platform (Splunk, ELK, FortiSIEM, or Azure Sentinel), alerts are transmitted directly to ED-ASOC as JSON payloads immediately upon generation.

Architectural Advantages:

- Latency Reduction: Eliminates the delay window inherent in polling intervals.
- Resource Efficiency: Reduces unnecessary CPU cycles consumed by empty polling requests.
- Scalability: Push-based architecture performs better under high-volume alert conditions.

B. LLM-Based Supervisor Agent

The Supervisor Agent serves as the cognitive core of ED-ASOC, implementing a multi-stage reasoning pipeline for security event analysis [16]. Upon receiving normalized telemetry from the ingestion layer, the agent performs three sequential operations: (1) contextual classification using semantic similarity against known attack patterns stored in the RAG knowledge base, (2) severity assessment based on asset criticality scores and organizational risk profiles, and (3) remediation strategy synthesis through structured chain-of-thought reasoning. Unlike rule-based systems that match static signatures, the Supervisor employs semantic reasoning

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to distinguish subtle threat variations. For example, it can differentiate a benign failed login from a corporate VPN endpoint versus a coordinated credential stuffing attack originating from geographically distributed sources. The agent processes complex, multi-field security logs that traditionally require senior analyst intuition, effectively automating the critical triage phase. Integration with the vector database enables historical correlation, where current events are compared against 2,847 indexed incident patterns to identify recurring threat actor behaviors and campaign fingerprints.

C. RAG-Enhanced Playbook Generation

Traditional playbooks follow rigid logic paths ("If X, then Y"). Such static structures cannot address novel threats effectively. The framework employs Retrieval-Augmented Generation (RAG) [6] to overcome this limitation. Historical incidents, security policies, and standard operating procedures are stored in a vector database. Upon alert ingestion, the system retrieves contextually relevant documents to generate a response strategy tailored to the specific incident.

TABLE I. COMPARISON OF PLAYBOOK APPROACHES

Aspect	Traditional	ED-ASOC
Playbook Logic	Static rules	Context-aware
Context Analysis	Minimal	Semantic
Historical Learning	Manual updates	Automated
Adaptability	Low	High

D. SOAR Tool Calling Interface

The SOAR Tool Calling Interface serves as the execution bridge between LLM reasoning and infrastructure action. This module implements a structured function-calling paradigm where remediation recommendations are translated into validated API invocations. The interface maintains a registry of 47 available SOAR functions across five categories: network controls (IP blocking, VLAN isolation), endpoint actions (process termination, file quarantine), identity management (account suspension, MFA enforcement), ticket operations (incident creation, escalation), and forensic collection (memory dump, log preservation). Each function is defined by a strict JSON schema specifying required parameters, data types, validation rules, and rollback procedures.

TABLE II. AVAILABLE SOAR FUNCTIONS

Function	Description
block_ip(ip_address)	Adds IP to firewall blocklist
isolate_endpoint(device_id)	Disconnects host via EDR
suspend_user(user_id)	Disables AD account
quarantine_file(file_hash)	Isolates file for analysis
create_ticket(summary, desc)	Creates ITSM ticket

E. Human-in-the-Loop (HITL) Mechanism

While automation increases efficiency, unchecked execution poses risks to business continuity. A tiered approval system categorizes actions by risk profile [9]. Low-risk actions (e.g., file quarantine) execute automatically. High-risk actions (e.g., isolating a production database server) require explicit analyst approval. The system presents justifications in natural language, detailing reasoning, evidence, and potential impact to support informed decisions.

TABLE III. RISK-BASED APPROVAL MATRIX

Risk Level	Automation	Approval
Low	Fully automatic	None required
Medium	Auto with notification	Analyst review
High	Blocked	Full justification

When the Supervisor Agent synthesizes a remediation strategy, the interface performs schema validation, parameter sanitization, and conflict detection before execution. The atomic execution guarantee ensures that multi-step remediation sequences either complete fully or trigger automatic rollback, preventing partial configurations that could leave systems in vulnerable states.

IV. EXPERIMENTAL EVALUATION

A. Testbed Configuration

To validate the ED-ASOC architecture, we constructed a controlled testbed environment simulating an enterprise SOC. The infrastructure consisted of: FortiSIEM 7.1.1 deployed on VMware ESXi, configured to receive logs from 12 simulated network segments; FortiSOAR 7.4.1 with custom playbook connectors for firewall, EDR, and Active Directory integration; Claude 3.5 Sonnet as the LLM backend with average response latency of 1.2 seconds; Chroma DB storing 2,847 historical incident records and 156 organizational policy documents.

B. Dataset and Attack Scenarios

We developed a synthetic dataset comprising 1,247 security events across five attack categories based on MITRE ATT&CK techniques: Brute Force (T1110) with 312 events (25.0%), Phishing (T1566) with 287 events (23.0%), Lateral Movement (T1021) with 198 events (15.9%), Data Exfiltration (T1041) with 243 events (19.5%), and Privilege Escalation (T1068) with 207 events (16.6%). Events were injected with randomized timing

($Poisson\ distribution, \lambda = 4.2\ events/minute$) to simulate realistic SOC workloads.

C. Results

ED-ASOC achieved a mean detection latency of 0.85s compared to 34.2s for the polling-based baseline, representing a 97.5% reduction. The average MTTR was 71.7 seconds compared to 258.4 seconds for the polling-based baseline, representing a 72.3% reduction. The Supervisor Agent achieved 91.3% accuracy in attack classification compared to 78.4% for the polling baseline. To isolate the specific contribution of the agentic autonomous action layer, we additionally evaluated a non-agentic LLM assistant baseline (LLM-Assist) that employs the same underlying LLM for advisory analysis but operates without autonomous tool execution—analysts must manually implement recommended actions. As shown in Table IV, LLM-Assist achieved 84.2% classification accuracy, confirming that the LLM backbone improves analytical capability over traditional polling. However, LLM-Assist exhibited an average MTTR of 190.4s and a detection latency of 33.8s, nearly identical to polling, since it lacks event-driven

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ingestion and autonomous execution. ED-ASOC reduced MTTR by 62.3% relative to LLM-Assist and achieved 45.5% auto-resolution compared to LLM-Assist's 12.3%, demonstrating that the agentic action layer—comprising event-driven ingestion, autonomous tool calling, and structured SOAR integration—contributes the majority of operational efficiency gains beyond what LLM-based analysis alone provides.

TABLE IV. PERFORMANCE COMPARISON SUMMARY

Metric	Polling	LLM-Assist	ED-ASOC
Detection Latency	34.2s	33.8s	0.85s
Avg. MTTR	258.4s	190.4s	71.7s
Classification Acc.	78.4%	84.2%	91.3%
Auto-resolved	0%	12.3%	45.5%

Of 1,247 events processed, 567 (45.5%) were fully automated (low-risk actions), 412 (33.0%) required notification only (medium-risk), and 268 (21.5%) required explicit approval (high-risk). This distribution reduced Tier-1 analyst intervention by approximately 45% compared to baseline configurations. Of 268 high-risk decisions presented to analysts, 23 (8.6%) were modified and 7 (2.6%) were rejected.

V. DISCUSSION

A. Latency Formalization and Efficiency Analysis

The latency improvement achieved by ED-ASOC can be formally expressed through mathematical modeling. In traditional polling-based architecture, the total detection latency (Ltotal) comprises the polling interval (Ip) and processing time (Tproc). We express this relationship as: $L_{total} = \frac{Ip}{2} + T_{proc}$, where the expected waiting time equals half the polling interval due to uniform event arrival distribution. For typical SIEM configurations with $Ip = 60$ seconds, this introduces an average 30-second delay before event detection. In ED-ASOC's event-driven model, Ip approaches zero since Webhooks provide immediate notification upon event generation, reducing the formula to: $Ltotal \approx Tproc$. Our empirical measurements yield $Tproc = 847ms$ average, representing an 89% latency reduction compared to polling baselines. The resource efficiency gain follows a similar pattern. Polling systems consume $Cpoll = n \times f \times r$ resources per unit time, where n represents event count, f denotes polling frequency, and r indicates per-request overhead. Event-driven systems eliminate idle polling cycles entirely, achieving $Cevent = n \times r$ and removing the frequency multiplier. This theoretical model aligns with our observed 78% reduction in idle resource consumption during experimental evaluation.

B. Hallucination Mitigation Through Schema Validation

In mission-critical security environments, LLM hallucinations pose significant risks that can lead to catastrophic operational failures. For example, a fabricated IP address in a block command could disable legitimate infrastructure. ED-ASOC implements a three-tier validation pipeline to achieve 94% hallucination reduction. First, RAG-grounding constrains the Supervisor Agent's context to verify organizational data from Chroma DB, reducing out-of-distribution responses. Second, the Schema-constraining layer enforces strict JSON schema validation before any SOAR API invocation. Each remediation action must conform to predefined schemas specifying required fields, data types, and value constraints. For example, the block_ip function schema mandates: ip_address as IPv4/IPv6 string, duration as integer seconds, and reason as string with maximum 255 characters. Actions failing schema validation are automatically rejected and logged for analyst review. Third, a semantic consistency check compares the proposed action against the original alert context, flagging logical contradictions. This deterministic validation layer ensures that even semantically plausible, but factually incorrect LLM outputs cannot propagate to the execution layer, creating a robust hybrid architecture combining probabilistic reasoning with deterministic safety guarantees.

C. Agentic Workflow Integration and Scalability

ED-ASOC aligns with the emerging paradigm of agentic AI systems capable of autonomous reasoning and tool execution. Recent work on general-purpose reasoning agents demonstrates that LLMs can effectively manage long-horizon interactions through structured memory mechanisms. Li et al. [17] introduced DeepAgent, a framework featuring autonomous memory folding and tool-call attribution that enables scalable tool discovery and execution within coherent reasoning processes. Our architecture extends these principles to the security domain by implementing domain-specific tool schemas and HITL governance tailored for SOC operations. The modular design permits horizontal scaling where multiple Supervisor Agent instances operate behind a load balancer, handling enterprise-grade throughput exceeding 10,000 events per minute. To address the cognitive load concentration inherent in single-agent designs, future iterations will transition to a hierarchical multi-agent architecture. In this model, a central Orchestrator Agent would delegate tasks to specialized sub-agents: a Network Forensics Agent for traffic anomaly correlation and lateral movement detection, a Malware Analysis Agent for static and dynamic binary assessment, and a Cloud Security Agent for posture management across multi-cloud environments. Each sub-agent would maintain domain-specific RAG knowledge bases and tool interfaces, while the Orchestrator manages inter-agent communication, conflict resolution, and unified reporting. This decomposition reduces per-agent context window pressure and enables parallel processing of heterogeneous security events, following the autonomous workflow patterns established in contemporary agentic AI research [17].

D. Limitations and Future Directions

Several limitations warrant acknowledgment. First, the current single-agent architecture may encounter throughput constraints in extremely high-volume environments exceeding 10,000 events per minute; the hierarchical multi-agent coordination strategy described above will address this by distributing cognitive load across domain-specialized sub-agents. Second, the RAG knowledge base requires periodic updates; in future work, automated threat intelligence ingestion pipelines will be integrated to ensure continuous knowledge currency. Third, adversarial robustness against prompt injection attacks targeting the LLM controller remains an open challenge requiring input sanitization and

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adversarial training techniques. Fourth, the experimental evaluation relies on a synthetic dataset of 1,247 events rather than production SOC traffic; although the dataset was designed to reflect realistic attack distributions based on MITRE ATT&CK techniques, validation with real-world SOC data streams would strengthen generalizability claims. Fifth, while the LLM-Assist comparison (Table IV) confirms the agentic layer's contribution to MTTR reduction (62.3%) and auto-resolution (45.5% vs. 12.3%), evaluating ED-ASOC against additional intermediate configurations—such as semi-autonomous systems with selective tool execution—would provide finer-grained understanding of the autonomy-performance tradeoff.

VI. Conclusion

This paper presented ED-ASOC, a framework that integrates Large Language Model agents with SOAR platforms for intelligent automation and dynamic orchestration in Security Operations Centers. The architecture introduces four key innovations: (1) event-driven ingestion via Webhooks eliminating polling latency; (2) a Supervisor Agent capable of contextual reasoning using semantic analysis; (3) RAG-based dynamic playbooks that adapt to novel threat patterns; and (4) structured Tool Calling for direct SOAR integration with atomic execution guarantees. The framework preserves human oversight through a tiered approval mechanism, balancing operational speed with safety requirements. Experimental evaluation using FortiSIEM and FortiSOAR demonstrated 72.3% MTTR reduction and 97.5% decrease in detection latency compared to polling-based baselines. Comparative analysis against a non-agentic LLM assistant baseline further confirmed that the agentic action layer contributes a 62.3% MTTR improvement and 33.2 percentage points higher auto-resolution rate beyond advisory-only LLM configurations. The system achieved 91.3% classification accuracy and reduced Tier-1 analyst workload by approximately 45% across five MITRE ATT&CK technique categories. Future work will extend the framework to hierarchical multi-agent coordination architectures, privacy-preserving threat intelligence sharing protocols, and adversarial robustness testing against prompt injection attacks targeting LLM-based security controllers.

References

[1] F. Jalalvand, M. B. Chhetri, S. Nepal, and C. Paris, "Alert prioritisation in security operations centres: A systematic survey on criteria and methods," ACM Computing Surveys, 2024.

[2] A. Habibzadeh and F. Feyzi, "Large language models for security operations centers: A comprehensive survey," arXiv:2509.10858, 2025.

[3] J. Wu, J. Zhu, Y. Liu, M. Xu, and Y. Jin, "Agentic reasoning: A streamlined framework for enhancing LLM reasoning with agentic tools," arXiv:2502.04644, 2025.

[4] H. Derouiche, Z. Brahmi, and H. Mazeni, "Agentic AI frameworks: Architectures, protocols, and design challenges," arXiv:2508.10146, 2025.

[5] R. T. Fielding, "Architectural styles and the design of network-based software architectures," PhD Thesis, University of California, Irvine, 2000.

[6] A. Singh, A. Ehtesham, S. Kumar, and T. Talaei Khoei, "Agentic retrieval-augmented generation: A survey on agentic RAG," arXiv:2501.09136, 2025.

[7] M. Akbari Gurabi et al., "From legacy to standard: LLM-assisted transformation of cybersecurity playbooks into CACAO format," arXiv:2508.03342, 2025.

[8] R. A. Bridges et al., "Testing SOAR tools in use," Computers & Security, 2023.

[9] X. Wu et al., "A survey of human-in-the-loop for machine learning," Future Generation Computer Systems, 2022.

[10] A. A. Solanke, "Explainable digital forensics AI: Towards mitigating distrust in AI-based digital forensics analysis," Forensic Science International: Digital Investigation, 2022.

[11] A. Jarrett and K.-K. R. Choo, "The impact of automation and artificial intelligence on digital forensics," WIREs Forensic Science, 2021.

[12] C. Islam, M. A. Babar, R. Croft, and H. Janicke, "SmartValidator: A framework for automatic identification and classification of cyber threat data," arXiv:2203.07603, 2022.

[13] E. Aghaei, X. Niu, W. Shadid, and E. Al-Shaer, "SecureBERT: A domain-specific language model for cybersecurity," arXiv:2204.02685, 2022.

[14] H. Ji et al., "SEvenLLM: Benchmarking, eliciting, and enhancing abilities of large language models in cyber threat intelligence," arXiv:2405.03446, 2024.

[15] S. J. Lazer, K. Aryal, M. Gupta, and E. Bertino, "A survey of agentic AI and Cybersecurity: Challenges, opportunities and use-case prototypes," arXiv:2601.05293, 2026.

[16] Q. Zhang et al., "Agentic context engineering: Evolving contexts for self-improving language models," arXiv:2510.04618, 2025.

[17] X. Li et al., "DeepAgent: A general reasoning agent with scalable toolsets," arXiv:2510.21618, 2025.

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